

Investigation of helium injection cooling to liquid oxygen propellant chamber

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Abstract

Sub-cooling of cryogenic propellant by helium injection is one of the most effective methods for suppressing bulk boiling and keeping sub-cooled liquid oxygen before rocket launch. In order to design the cooling system, understanding of the limitations of heat and mass transfer is required. In this paper, an analytical model for the helium injection system is presented. This model's main feature is the representation of bubbling system using finite-rate heat transfer and instantaneous mass transfer concept. With this simplified approach, the effect of helium injection to liquid oxygen system under several circumstances is examined. Experimental results along with simulations of single bubble rising in liquid oxygen and bubbling system are presented with various helium injection flow rates, helium temperatures, and injection methods. The overall cooling effect for rocket application is also discussed.

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1. Introduction

One of the primary objectives of a rocket propellant feeding system is to supply propellant to the engine at the required temperature and pressure. Numerous attempts have been made to provide proper cryogenic propellant temperature required for engine ignition while preventing geysering in the propellant feedline. These methods relied on sub-cooling of cryogenic propellants. Recently, sub-cooling of cryogenic propellant have been done not only to meet temperature requirements but also to improve the performance of launch vehicles by reducing lift-off weight [1]. Among traditional methods, sub-cooling of cryogenic propellant by helium injection is known to be

one of the most effective methods for suppressing bulk boiling and keeping liquid oxygen sub-cooled before rocket launch. A similar concept is also applied for ground pre-pressurization for upper stages [2].

The fundamental principle of sub-cooling of cryogenic propellant by helium injection is diffusional mass transfer. Since the injected helium is pure, the difference in the partial pressure of the oxygen in the helium bubble and the vapor pressure of the LOX causes diffusional mass transfer of oxygen into the helium gas. This diffusion process (or boiling process) results in a cooling effect caused by the absorption of vaporization heat from the surrounding liquid oxygen [3].

Previous studies focused on obtaining possible maximum sub-cooling assuming the injected helium and the oxygen vapor to be in phase equilibrium [4,5]. However, in reality, the phase equilibrium assumption is limited by finite rate of heat and mass transfer. In order to design efficient cooling systems, we need to understand the limitations of heat and mass transfer rate.

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Nomenclature

A, A_s	surface area (m ²)	V	volume (m ³)
Bi	Biot number	v	diffusion volume in Eq. (13)
C_D	drag coefficient of bubble	$V_{b,i}$	volume of detached bubble (m ³)
C_p	heat capacity (kJ/kg K)	X	mole fraction ratio defined in Eq. (5)
D, D_b	bubble diameter (mm)	$x_{O_2,u}$	oxygen mole fraction at 'u' surface
$D_{b,c}$	correlated bubble diameter in Eq. (29) (mm)	$x_{O_2,s}$	oxygen mole fraction at 's' surface
$D_{b,i}$	detached bubble diameter at orifice (mm)	Z	compressibility factor
D_{O_2-He}	diffusion coefficient for oxygen flowing through helium (m ² /s)	<i>Gradients and differences</i>	
h	local heat transfer coefficient (W/m ² K)	ΔT_{excess}	temperature difference defined in Eq. (25)
$\bar{h}$	average heat transfer coefficient (W/m ² K)	<i>Greek letters</i>	
h_g	enthalpy of gas (kJ/kg)	α	void fraction
h_l	enthalpy of liquid (kJ/kg)	μ	dynamic viscosity (Pa s)
h_{fg}	heat of vaporization (kJ/kg)	μ_w	dynamic viscosity of liquid at surface temperature (Pa s)
i	real interface	ν_v	kinematic viscosity of vapor (m ² /s)
k	thermal conductivity (W/m K)	ρ	density (kg/m ³)
k_v	thermal conductivity of oxygen vapor (W/m K)	σ	surface tension of liquid (N/m)
K	non-dimensional parameter defined in Eq. (15)	<i>Subscripts</i>	
m	mass (kg)	amb	ambient
$\dot{m}$	mass flow rate (kg/s)	b	bubble
$\dot{m}_{v,h}$	evaporation rate due to heat flux (kg/s)	conv	convection
$\dot{m}_{v,\text{diff}}$	evaporation rate due to diffusion (kg/s)	fb	flow boiling
$\dot{m}_{\text{ev}}$	total evaporation rate ($\dot{m}_{v,h} + \dot{m}_{v,\text{diff}}$) (kg/s)	g	gas
$\dot{m}_{\text{ev}}''$	total evaporation rate per area (kg/s m ²)	gi	gas-to-interface
M	molecular weight (kg/mol)	GHe	helium gas
M_b	molecular weight of mixture in bubble (kg)	GOX	oxygen gas
n	mole	i	real interface
$\overline{Nu}_D$	average Nusselt number	l	liquid
P	pressure (MPa)	li	liquid-to-interface
P_{GHe}	gaseous helium partial pressure (MPa)	o	orifice
P_{GOX}	gaseous oxygen partial pressure (MPa)	pb	pool boiling
P_{O_2}	oxygen partial pressure (MPa)	s	vapor side interface
$P_{GOX,i}$	gaseous oxygen partial pressure at interface (MPa)	sat	saturation
$P_{GOX,g}$	gaseous oxygen partial pressure in gas (bubble) region (MPa)	sc	single phase convection
$P_{\text{sat},s}$	saturation pressure at 's' surface (MPa)	sys	system
Pr	Prandtl number	u	liquid side interface
q''	heat flux (W/m ²)	v	oxygen vapor
$\dot{Q}$	heating rate (W)	w	wall
$\dot{Q}_g$	volumetric gas flow rate through orifice (m ³ /s)	<i>Abbreviations</i>	
R	universal gas constant (kJ/kmol K)	atm	atmosphere
R_{GHe}	gas constant of helium (kJ/kg K)	GHe	gaseous helium
R_{GOX}	gas constant of oxygen (kJ/kg K)	GOX	gaseous oxygen
r_b	bubble radius (mm)	LOX	liquid oxygen
R_o	orifice radius (mm)	min	minute
Ra	Rayleigh number	TT	temperature transmitter
Re	Reynolds number	PT	pressure transmitter
Sc	Schmidt number	LT	level transmitter
T	temperature (K)	FM	flow meter
u_b	bubble rising velocity (m/s)		
u_g	gas injection velocity (m/s)		

In this paper, an analytical model for the helium injection system is presented. This model's main feature is the representation of the bubbling system using finite rate heat transfer and instantaneous mass transfer concept. With this simplified approach, the effects of helium injection under several circumstances are studied. Experimental results along with simulations of various injection flow rates for bubbling, helium temperatures and injection methods are presented in this paper.

2. Theoretical investigation

2.1. Principle of sub-cooling by helium injection

The principle of sub-cooling by helium injection is portrayed in Fig. 1. In order to clearly define partial pressure at the interface surface, imaginary surfaces denoted by 'u' and 's' are introduced, on both sides of the real interface, each indefinitely close to the real interface (i), for liquid oxygen (LOX) evaporating into helium gas stream [6]. If we ignore the small amount of helium gas dissolved in the LOX, the oxygen mole fraction at the 'u' surface $x_{O_2,u}$ is 1. Except for extreme circumstances, the mixture of oxygen vapor and helium gas at the 's' surface must be in thermodynamic equilibrium with the LOX at the 'u' surface. The oxygen mole fraction $x_{O_2,s}$ is, therefore, determined by assuming that the partial pressure $P_{O_2,s}$ is same as the saturation vapor pressure of oxygen at the liquid temperature. If the diffusional mass transfer effect is excluded, the heat transfer from the bulk gas to the real surface $\dot{Q}_{gi}$ is nearly equal to the product of evaporation rate due to heat transfer $\dot{m}_{v,h}$ and heat of vaporization h_{fg} as shown in Eq. (1). In this case, liquid oxygen is not cooled, which can occur if oxygen gas instead of helium is injected into the LOX.

$$\dot{Q}_{gi} \approx \dot{m}_{v,h} h_{fg} \quad (1)$$

However, if pure helium is injected into liquid LOX, the partial pressure of the oxygen vapor at the 's' surface would significantly differ from its bulk gas partial pressure, which

causes diffusional mass transfer of oxygen according to Fick's law of diffusion from the 's' surface. In this case to satisfy the saturation condition at surface 's', more liquid oxygen should be vaporized and supplied to the 's' surface as in Eq. (2), where $\dot{m}_{v,h}$ and $\dot{m}_{v,diff}$ are the LOX evaporation rate due to heat transfer and mass diffusion, respectively. The added vaporization of oxygen due to mass diffusion during helium injection removes heat from the liquid oxygen and causes the bulk liquid temperature to decrease with time according to Eq. (3). In Eq. (3), $d(m_{GHe} C_{p,GHe} T_{GHe})/dt$ term does not contribute cooling of the LOX if injected helium temperature is higher than LOX temperature, whereas, as $\dot{m}_{v,diff}$ is increased, greater sub-cooling can be obtained. Therefore, the diffusional mass transfer is the main drive of LOX sub-cooling, and the LOX sub-cooling by helium injection can be characterized by diffusion-driven evaporative cooling.

$$\dot{m}_{ev} = \dot{m}_{v,h} + \dot{m}_{v,diff} \quad (2)$$

$$m_{LOX} C_{p,LOX} \frac{dT_{LOX}}{dt} = -\dot{m}_{ev} h_{fg} - \frac{d(m_{GHe} C_{p,GHe} T_{GHe})}{dt} + \dot{Q}_{amb} \quad (3)$$

2.2. Investigations of limiting cases

In the investigation of helium injection cooling, three limiting cases can be established in terms of heat and mass transfer aspects. The first case is instantaneous heat and mass transfer. In this case, the heat and mass transfer rate is infinite. The bulk liquid and gas are always in equilibrium state. This case can be characterized by a so-called equilibrium bubbler.

The second case is finite heat transfer and instantaneous diffusional mass transfer. In this case, the temperatures of the bulk liquid and gas are different due to finite heat transfer. However due to the infinite mass transfer rate, the partial pressure of the oxygen vapor at the 's' surface is equal to the bulk gas partial pressure of oxygen. This case can be characterized by a thermally non-equilibrium bubbler.

The third case is more general; finite heat and mass transfer. This case is certainly the most realistic description of the phenomenon. However, the analysis involves many uncertain transport properties and complex calculation procedures. Due to the complexity of the third case, the first or the second cases are usually applied under proper circumstances. The mathematical approaches for dealing with each case are reviewed and presented in the following sections.

2.2.1. Instantaneous heat and mass transfer (equilibrium bubbler)

In this case, the vaporization time, which is determined by heat and mass diffusion, is extremely small compared with the helium gas holdup. It is assumed that helium bubbles become completely saturated instantaneously. The equilibrium point is established when the partial pres-

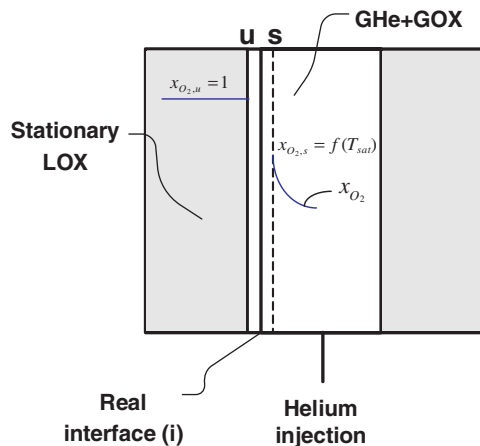


Fig. 1. Schematics of principle on helium injection cooling.

sure of oxygen P_{GOX} in bubble is exactly equal to the saturated vapor pressure of the liquid oxygen at the particular temperature of the liquid. According to the real gas state law in Eq. (4), the mole fraction ratio X can be expressed as in Eq. (5).

$$PV = ZnRT \quad (4)$$

$$X = \frac{n_{GHe}}{n_{GOX}} = \frac{P_{GHe}}{P_{GOX}} \cdot \frac{Z_{GOX}(P, T)}{Z_{GHe}(P, T)} \quad (5)$$

If the system pressure P is maintained constant during helium injection, the helium partial pressure P_{GHe} must obey the following equation.

$$P_{GHe} = P - P_{GOX} \quad (6)$$

Differentiating Eq. (5) results in

$$\frac{\dot{n}_{GOX}}{\dot{n}_{GHe}} = \frac{1}{X} - \frac{\dot{X}}{X^2} \cdot \frac{n_{GHe}}{\dot{n}_{GHe}} \quad (7)$$

Assuming that no mass exchange takes place between the rising gas phase and the surrounding liquid subsequent to the initial instantaneous equilibrium, attained at the moment of injection, Eq. (7) reduces to Eq. (8) [4].

$$X = \frac{\dot{n}_{GHe}}{\dot{n}_{GOX}} = \frac{\dot{m}_{GHe} R_{GOX}}{\dot{m}_{GOX} R_{GHe}} \quad (8)$$

The value P_{GOX} ($P_{GOX} = P_{sat}(T_{LOX})$) can be found from the liquid oxygen saturation curve as in Fig. 2, and P_{GHe} is obtained from Eq. (6). Substituting for the partial pressures in Eq. (5), the mole fraction ratio X can be expressed as a function of LOX temperature. From Eq. (8), the oxygen vaporization rate ($\dot{m}_{GOX}$) can be found from the supplied helium flow rate ($\dot{m}_{GHe}$) and X , and finally with Eq. (3), LOX sub-cooling can be determined. This approach theoretically gives the maximum LOX evaporation rate and sub-cooling rate.

Preliminary analysis is carried out to account for the instantaneous heat and mass transfer case. The analysis conditions are as follows. There are 90 l of liquid oxygen (saturated at 1 atm) and 288 K helium gas is injected dur-

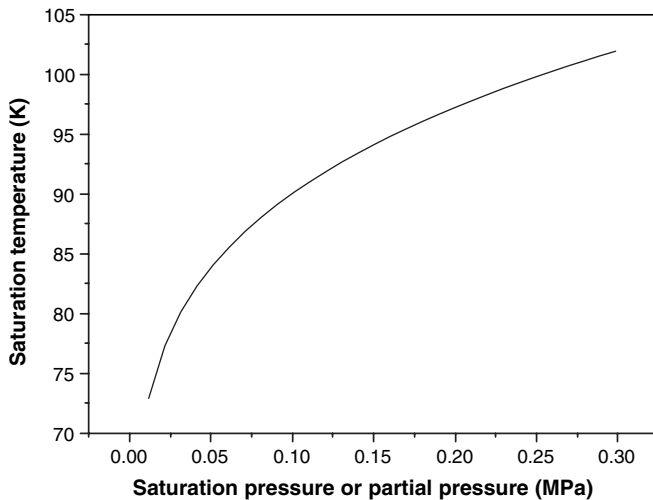


Fig. 2. Oxygen saturation curve.

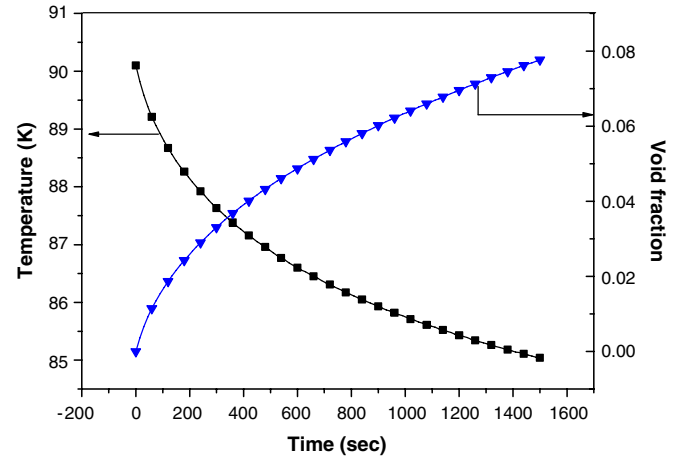


Fig. 3. Temperature and void fraction history (infinite heat transfer/instantaneous mixing case).

ing 25 min into liquid oxygen. The helium injection rate is specified to be 90 l/min ($\dot{V}_{GHe}/V_{LOX} = 1 \text{ min}^{-1}$). During the helium injection process, the liquid oxygen is assumed to maintain 1 atm via atmospheric venting and the heat flux from ambient sources is ignored. Analysis results are presented in Fig. 3. Liquid oxygen is sub-cooled down to 84.5 K in this case. The void fraction is calculated based on the specified helium injection rate and the vaporized LOX calculated from Eq. (8). It is increased up to 0.08.

2.2.2. Finite heat transfer and instantaneous diffusion mass transfer (thermally non-equilibrium bubbler)

In this case, the liquid and the gas are not at the same temperature due to finite heat transfer, while the concentration is equal throughout the gas region due to the infinite rate diffusional mass transfer. The mixing of oxygen vapor and helium gas is assumed to occur in a negligible time compared with heat transfer. The surface energy balance diagram associated with simultaneous heat and mass transfer is shown in Fig. 4. If the steady energy flow equation is applied to the control volume located between the ‘u’ and

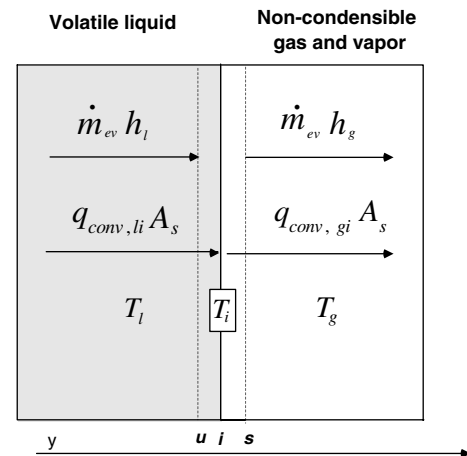


Fig. 4. Schematic diagram for surface energy balance.

's' surfaces, with radiation heat flux ignored, the energy balance can be expressed as in Eq. (9). It shows the evaporation rate $\dot{m}_{ev}$ can be calculated with heat transfer coefficients (gas-to-interface: h_{gi} , liquid-to-interface: h_{li}), surface area (A_s) and interface temperature (T_i) for a given system [6].

$$\dot{m}_{ev} = A_s(h_{gi}(T_g - T_i) - h_{li}(T_i - T_l))/h_{fg} \quad (9)$$

If instantaneous mixing is assumed, the gradient of partial pressure of oxygen at the 's' surface is zero, which means the partial pressure at the 's' surface $P_{GOX,s}$ is equal to the partial pressure of the bulk gas region $P_{GOX,g}$ as in Eq. (10).

$$P_{GOX,s} = P_{GOX,g} \quad (10)$$

As the temperature is continuous across the phase interface and the 's' surface is assumed to be indefinitely close to the interface i, if thermodynamic equilibrium of oxygen vapor and helium gas at the interface is applied, T_i in Eq. (9) can be calculated based on the partial pressure of oxygen in the gas region as in Eq. (11) using the oxygen saturation curve.

$$T_i = T_{sat}(P_{GOX,g}) \quad (11)$$

A temperature and partial pressure profile is schematically illustrated in Fig. 5 in case of the finite heat transfer and instantaneous mixing.

From the macroscopic point of view, as the helium injection rate increases, the oxygen partial pressure in the gas region becomes lower, which leads to lower T_i according to the oxygen saturation curve as shown in Fig. 2. In Eq. (9), as T_i decreases, the vaporized mass flow rate $\dot{m}_{ev}$ increases. As the interface temperature T_i depends on the partial pressure of oxygen $P_{GOX,g}$ in the gas region, lowering $P_{GOX,g}$ is advantageous for evaporative cooling purpose. This can be carried out by increasing the helium injection flow rate or decreasing the amount of vaporized oxygen due to heat flux which depends on the injected helium temperature.

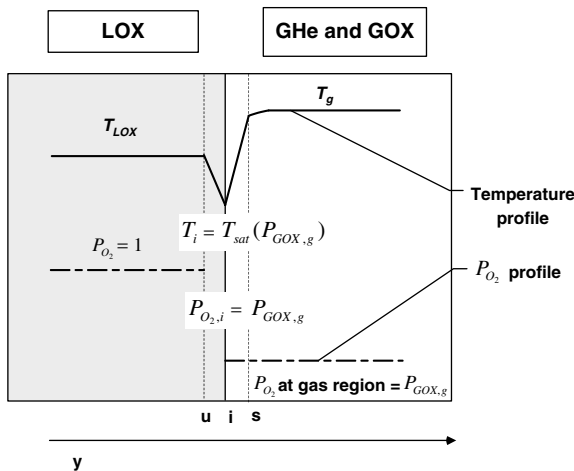


Fig. 5. Schematic diagram of temperature and partial pressure profile (finite heat transfer/instantaneous mixing case).

2.2.3. Finite heat transfer and finite diffusion mass transfer

In this case, mass diffusion is also a limiting factor at the primary interface. Usually, the interface temperature is not at saturation state. It is shifted slightly due to diffusion effects. $\dot{m}_{ev}$, T_i , h_{gi} are iteratively calculated with Eqs. (9), (12) and (13) [7].

$$\frac{h_{gi}}{\dot{m}_{ev}/(\rho_{LOX,l} - \rho_{LOX,s})} = \left[\frac{P_{sys}}{P_{GHe,g}} \cdot \rho_{GHe,g} \cdot \frac{\ln\left(\frac{\rho_{GHe,s}}{\rho_{GHe,g}}\right)}{\rho_{GHe,g} - \rho_{GHe,s}} \right]^{-1} \times \left[\rho_g \cdot C_{p,g} \cdot \left(\frac{k_g}{D_{O_2-He}} \right)^2 \right]^{1/3} \quad (12)$$

$$D_{O_2-He} = \frac{10^{-3} \cdot T^{1.75} \cdot [(M_{O_2} + M_{He})/(M_{O_2}M_{He})]^2}{P \cdot [(\sum v)_{O_2}^{1/3} + (\sum v)_{He}^{1/3}]^2} \quad (13)$$

where D_{O_2-He} is the diffusion coefficient for oxygen flowing through helium, M is molecular weight and v is diffusion volume, respectively [8].

2.2.4. Selection of analysis model

In terms of heat transfer, thermal equilibrium can be assumed if helium bubbles injected at ambient temperature (288 K) reach the LOX temperature before they enter into the ullage gas region. For long pipelines such as those used in the first stage of liquid rocket propulsion systems, the bubbles have a longer residence time and therefore they have a greater chance to reach thermal equilibrium. However, in the upper stage pipes or relatively small tanks, the bubbles have less chance to reach thermal equilibrium which leads to non-equilibrium between the bulk liquid and bubbles.

The applicability of the instantaneous diffusion assumption can be deduced from a heat and mass transfer analogy. The Biot number condition ($Bi = h(V/A)/k < 0.1$) indicates that the resistance to conduction is much smaller than the resistance of the thermal boundary layer surrounding the body, so a lumped heat capacity model may be used. Using the heat and mass transfer analogy in Eq. (14), thermal conductivity k in the Biot number equation can be replaced and applied to a spherical single bubble as in Eq. (15). In Eq. (15), the dimensionless parameter K is the modified the Biot number which can be a criterion determining whether the diffusion resistance can be ignored and the gas space can be regarded as perfectly mixed one.

$$Sc = Pr, \quad \frac{\mu}{\rho D_{O_2-He}} = \frac{C_p \mu}{k} \quad (14)$$

$$K = \frac{h_{lu,b}(V_b/A_b)}{\rho_b C_{p,b} D_{O_2-He}} < 0.1 \quad (15)$$

In this paper, the LOX tank of the helium injected system is 2.5 m long and a similar shape to typical upper stage pipelines or propellant tanks. In the preliminary experiments, the measured temperature of the ullage increased during helium injection which indicated thermal non-equilibrium situation between the LOX and the bubbles. The

dimensionless parameter K , calculated from Eq. (15) is less than 0.1. Therefore, the assumption of instantaneous diffusion is applied. The calculation procedures for K is deeply discussed in a later section.

3. Single bubble analysis

For cooling purpose, it is advantageous to increase the overall bubble surface area. As the bubble population increases, the bubbles collide and are merged, which decreases the bubble surface area abruptly. It is desirable to limit bubble population to avoid such an interaction. Wallis [9] observed that for a volume fraction of gas less than 10%, the rate of collision and agglomeration of bubbles is usually slow, and for that gas volume fraction range (void fraction α) the bubbles can be treated independently from other bubbles. From this viewpoint, one may say that single bubble characteristics can be extended to account for the whole bubbling system without significant error for that range of gas volume.

Because the void fraction at maximum sub-cooling condition is below 0.1 as shown in Fig. 3, the average bubble surface area and heat transfer coefficients during bubble residence time in liquid oxygen can be obtained from single bubble analysis. Single bubbles are formed at the orifice. They detach and grow due to mass transfer, or they contract due to heat transfer with cold liquid. For the bubble analysis, the following reference conditions are established. Saturated liquid oxygen (90 l) is filled and 90 l/min, 288 K helium gas is injected into the liquid oxygen. During the helium injection, the liquid oxygen is assumed to maintain 1 bar by atmospheric venting. The injection orifice diameter is 6.3 mm. Analysis is carried out with various helium injection flow rates.

3.1. Bubble formation at orifice

Under normal conditions, there are three known bubbling regimes at the submerged orifice. These regimes are the static, the dynamic, and the turbulent [10]. The static regime occurs at very low gas flow rates when both inertial and viscous forces may be overlooked. This limiting configuration is established under the combined action of surface tension and gravity force, and the regime ceases to be valid when the orifice radius R_o is comparable with the bubble radius, that is,

$$R_o > 0.5 \left[\frac{\sigma}{g(\rho_l - \rho_g)} \right]^{1/2} \quad (16)$$

The turbulent regime occurs at very high flow rates when successively emerged bubbles coalesce close to the orifice, and after being disintegrated into smaller bubbles of various sizes and irregular shapes, they then participate in a vigorous swirling motion. Kutateladze and Styrikovich [11] presented the condition for the bubble jet as in Eq. (17).

$$\frac{u_g \sqrt{\rho_g}}{[g\sigma(\rho_l - \rho_g)]^{1/4}} > 1.25 \left[\frac{\sigma}{g(\rho_l - \rho_g)R_o} \right]^{1/2} \quad (17)$$

The dynamic regime is characteristic of systems using an intermediate range of gas flow rate, that is, the flow rates which are most often used in industrial applications [10]. The main feature of the dynamic regime is that the bubbles may be considered more or less independently from one another. For the dynamic regime, Davidson and Schuller [12] have made calculations for viscous liquids and obtained the results as in Eq. (18).

$$V_{b,i} = \left(\frac{4\pi}{3} \right)^{1/4} \left[\frac{15\mu_l Q_g}{2g(\rho_l - \rho_g)} \right]^{3/4} \quad (18)$$

where $V_{b,i}$ is the detached bubble volume and Q_g is the volumetric gas flow rate through the orifice, respectively. At the reference analysis condition of this paper, Eq. (16) is satisfied and Eq. (17) is not satisfied, so Eq. (18) is applied for estimating the detached bubble size assuming a spherical bubble.

3.2. Bubble growth

The detached bubble may grow due to oxygen vapor transfer across the bubble surface, or may shrink as the bubble gas temperature is lowered as the bubble rises. For simplicity, it is assumed that the initial bubble shape is spherical and it retains its original shape throughout its trip through the liquid oxygen column. The pressure in a bubble can be expressed as the summation of the surrounding liquid's pressure at the surface (P_l) and the contribution of the surface tension as in Eq. (19)

$$P = P_l + \frac{2\sigma}{r_b} \quad (19)$$

Differentiating Eq. (19), (20) is obtained

$$\frac{dP}{dt} = -\frac{2\sigma}{r_b^2} \frac{dr_b}{dt} \quad (20)$$

In the bubble the real gas state equation can be applied as in Eq. (21).

$$PV = ZnRT = Z \frac{m_b}{M_b} RT \quad (21)$$

where m_b is the bubble mass and M_b is the molecular weight of oxygen vapor and helium mixture at the specific time, respectively. Differentiating the both sides of Eq. (21) and substituting the value (dP/dt) from Eq. (20), the bubble growth rate can be expressed as in Eq. (22).

$$\frac{dr_b}{dt} = \frac{\dot{m}_b \frac{zRT}{M_b} + m_b \frac{d}{dt} \left(\frac{m_b RT}{M_b} \right)}{-\frac{8\pi\sigma r_b}{3} + 4\pi r_b^2 P} \quad (22)$$

The first term of the RHS in the numerator indicates the bubble growth due to mass diffusion of oxygen vapor, whereas the second term of the RHS in the numerator

means the bubble radius change due to temperature decrease by heat transfer with cryogenic liquid oxygen.

3.3. Bubble rising velocity

After detachment from the orifice, a bubble rapidly accelerates to its terminal velocity u_b . The value of u_b is determined by the balance between the buoyant rise force and the drag force. The rise velocity of a single bubble inherently depends on its sizes [13]. For bubbles with diameters greater than 5 mm, Haberman and Morton [14] proposed the drag coefficient and the bubble terminal velocity as in Eq. (23).

$$C_D = 2.6, \quad u_b = 1.05 \left[\frac{g(\rho_l - \rho_g)}{\rho_l} \cdot \left(\frac{D_b}{2} \right) \right]^{1/2} \quad (23)$$

Considering the initial bubble size obtained from Eq. (18), Eq. (23) is used for estimating bubble rising velocity.

3.4. Heat transfer at interface

The heat transfer coefficients that are applied to the main interface in surface energy balance equation (Eq. (9)) are composed of two components (h_{gi} and h_{li}). h_{gi} is the gas-to-interface heat transfer coefficient and h_{li} is the liquid-to-interface heat transfer coefficient, respectively. h_{gi} is the single phase convective heat transfer coefficient. For large bubbles (>2.5 mm diameter), natural convection inside a spherical cavity of diameter D was correlated by Kreith [15] using the empirical relation as in Eq. (24).

$$\overline{Nu}_D = 0.13 Ra_D^{1/4}, \quad 10^9 \leq Ra_D \leq 10^{12} \quad (24)$$

The reference condition satisfies the Ra range in Eq. (24) and is applied for the bubble gas-to-interface heat transfer coefficient (h_{gi}) estimation.

h_{li} is the boiling heat transfer coefficient associated with LOX vaporization in the ‘u–i’ region as shown in Fig. 4. When volatile liquid is in contact with a hot solid wall, ΔT_{excess} ($T_w - T_{sat}$), which is defined as the temperature difference between a solid wall temperature (T_w) and the saturation temperature of liquid (T_{sat}), is generally used for heat transfer correlations. For bubble problems, the rising bubble is in contact with liquid instead of solid wall. Even though a large number of studies have been made on boiling heat transfer between a solid wall and volatile liquid, little is known about boiling heat transfer between a volatile liquid and gas. Observing related surface temperatures for the two cases, between which two phase film is formed, the solid wall temperature is replaced by the bulk liquid oxygen temperature (T_{LOX}) as illustrated in Fig. 6. And if instantaneous diffusion is assumed, the saturation temperature at the interface may be calculated from the partial pressure of liquid oxygen in the inner bubble region. The local temperature varies from T_{LOX} to $T_{sat}(P_{GOX,g})$ across the ‘u–i’ region as shown in Fig. 5.

In the case of bubbles, the excess temperature may be defined as the temperature difference between the bulk

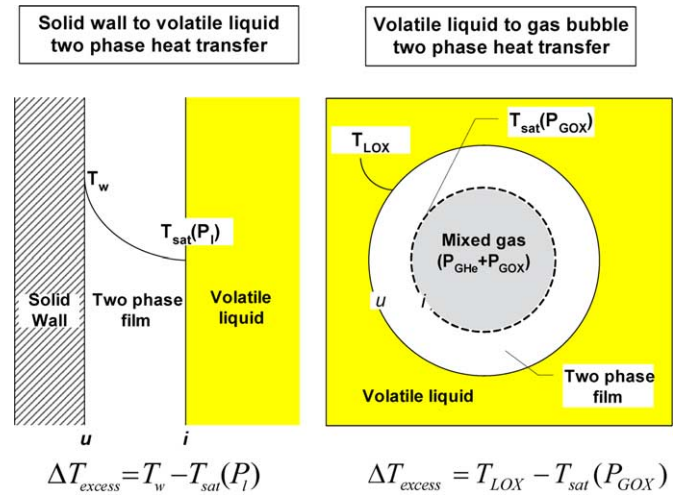


Fig. 6. Comparison of ΔT_{excess} definition.

liquid oxygen temperature independent of system pressure and the interface temperature corresponding to the partial pressure of oxygen in the gas region as in Eq. (25). The analogy of ΔT_{excess} for the two cases is described in Fig. 6.

$$\Delta T_{excess} = T_{LOX} - T_{sat}(P_{GOX,g}) \quad (25)$$

For high ΔT_{excess} , film boiling is usually dominant. Based on the experimental data for liquid nitrogen, Frederking and Clark [16] presented the pool boiling heat transfer correlation associated with film boiling as following equation.

$$\bar{h} = 0.15 \left[\frac{(\rho_l - \rho_v) g h'_{fg} k_v^2}{v_v \Delta T_{excess}} \right]^{1/3} \quad (26)$$

$$\frac{D^3 (\rho_l - \rho_v) g h'_{fg}}{k_v v_v \Delta T_{excess}} > 5 \times 10^7$$

where $h'_{fg} = h_{fg} + 0.5 C_{p,v} \Delta T_{excess}$. The subscript l and v are defined as liquid property and vapor property of volatile liquid, respectively. Considering the fact that 288 K helium is injected into cryogenic liquid oxygen, we use Eq. (26) for the liquid-to-interface film boiling heat transfer (h_{li}) estimation using ΔT_{excess} defined in Eq. (25). In Eq. (26), h_{fg} , ρ_l and σ are to be evaluated at T_{LOX} ; all other properties should be evaluated at the mean film temperature.

In fact, Eq. (26) is pool boiling correlation under the assumption that the bulk liquid is stationary. Since the bubble rises with its terminal velocity in the LOX tank, the flow boiling effect may need to be considered. There is no general definitive heat transfer correlation for flow of boiling bubbles. We use one of the earliest and simplest methods that Rohsenow [17] presented the experimental data on nucleate and film boiling with convection. They are adequately correlated by the additive formula as in Eq. (27).

$$q''_{fb} = q''_{pb} + q''_{sc} \quad (27)$$

In the above expression, q''_{fb} is the flow boiling heat flux, q''_{pb} is the pool boiling heat flux calculated based on Eq. (26) and q''_{sc} is the single phase convection heat flux in liquid.

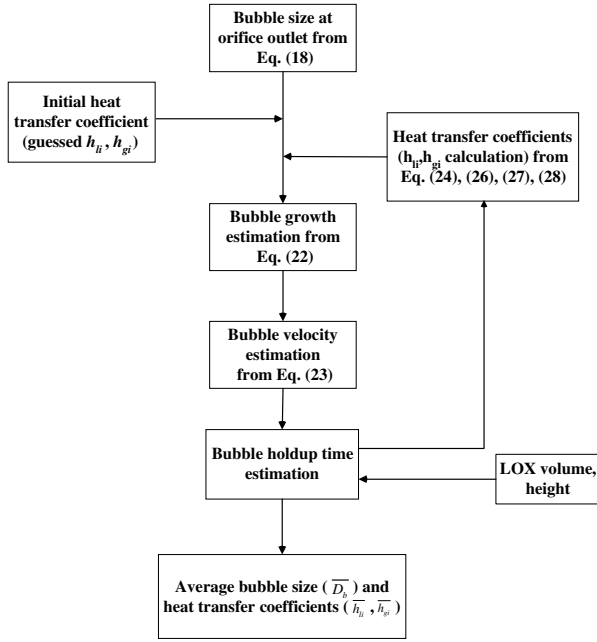


Fig. 7. Single bubble analysis procedure.

For an average single phase heat transfer coefficient of liquid on a spherical surface Whitaker's correlation [18] is used as shown in Eq. (28)

$$\overline{Nu}_D = 2 + (0.4Re_D^{1/2} + 0.06Re_D^{2/3})Pr^{0.4} \left(\frac{\mu_l}{\mu_w} \right)^{1/4} \quad (28)$$

All the physical properties in Eq. (28) are evaluated at the bulk liquid oxygen temperature T_{LOX} , except μ_w , which is the viscosity evaluated at the surface temperature. For surface temperature, the saturation temperature corresponding to the partial pressure of oxygen in the gas region $T_{sat}(P_{GOX,g})$ is applied.

3.5. Bubble analysis procedures

Calculation procedures in the bubble analysis are summarized as shown in Fig. 7. First, the bubble detachment size is calculated. And then with initial heat transfer coeffi-

cients, the bubble growth and the bubble rising velocity are estimated. With given LOX volume, the bubble residence time is calculated. During the calculated bubble residence time, the average values of heat transfer coefficients, the bubble size and the bubble rising velocity are re-calculated. The procedures are repeated until the calculated values converge. Finally, the average heat transfer coefficients, and the average bubble size during residence time are obtained.

4. Bubbling system analysis

Table 1 shows the input conditions for bubbling system analysis. The parameter values are obtained from single bubble analysis discussed in the previous section. The reference analysis conditions are established: 90 l of saturated liquid oxygen (saturated at 1 atm) is filled. And 90 l/min ($\dot{V}_{GHe}/V_{LOX} = 1 \text{ min}^{-1}$), 288 K helium gas is injected into liquid oxygen during 25 min. During helium injection the liquid oxygen is assumed to maintain 1 bar via atmospheric venting and the heat flux from ambient sources is ignored. The injection orifice diameter is 6.3 mm. As shown in Table 1, the calculated K number (Eq. (15)), which determines whether the diffusion resistance can be ignored and the gas space can be regarded as perfectly mixed, is less than 0.1. Therefore, the assumption of finite heat transfer and instantaneous diffusion mass transfer is confirmed and applied. Individual bubbles experience heat and mass transfer during the estimated residence time and are vented to atmosphere as shown in Fig. 8. During bubble system analysis, if the volume fraction of gas exceeds 0.1, the following bubble diameter correlation is applied [19].

$$D_{b,c} = D_b \left(\frac{9\alpha}{1-\alpha} \right)^{1/3} \quad \text{for } \alpha \geq 0.1 \quad (29)$$

where α is the volume fraction of the gas, and $D_{b,c}$ is the correlated bubble size, D_b is the instantaneous bubble size obtained from calculations in the preceding section, respectively.

Fig. 9 shows the GOX partial pressure profile and the LOX saturation pressure profile. Due to finite heat trans-

Table 1
Input conditions for bubbling system analysis

Parameter	Value	Reference
Detached bubble diameter ($D_{b,i}$)	5.1 mm	
Average bubble diameter ($\bar{D}_b$)	4.5 mm	
Average bubble volume ($\bar{V}_b$)	$4/3 \times \pi \times (\bar{D}_b/2)^3$	
Number of bubble (N_b)	$V_g/\bar{V}_b$	Total gas volume (V_g) depends on helium injection rate
Overall bubble surface area ($\sum A_b$)	$N_b \times 4 \times \pi \times (\bar{D}_b/2)^2$	
System pressure	1 atm	System pressure is maintained by atmospheric venting
Injected helium temperature	288 K	
Helium injection rate	0.0015 m ³ /s	$\dot{V}_{He}/V_{LOX} = 1.39 \text{ min}^{-1}$
LOX temperature	90.2 K	
h_{gi}	40 W/m ² K	
h_{hi}	260 W/m ² K	
K (Eq. (15))	0.08	$K < 0.1$

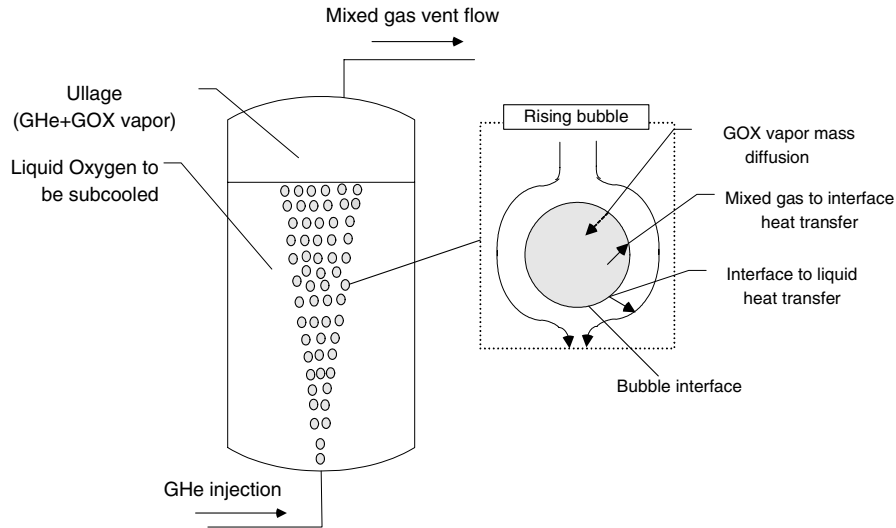


Fig. 8. Schematic diagram of bubbling system analysis.

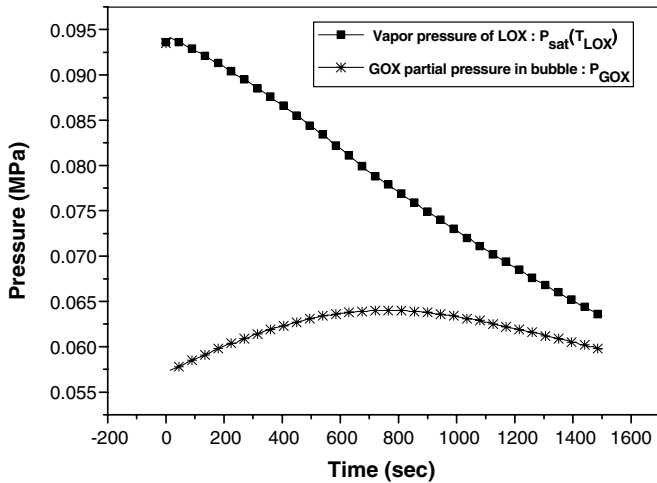


Fig. 9. GOX partial pressure and LOX saturation pressure profile (finite heat transfer/instantaneous mixing case).

fer, the GOX partial pressure is lower than LOX saturation pressure corresponding to bulk LOX temperature.

It is worthwhile to peruse the vaporization rate of the LOX. We could identify the vaporization sources into two components as in Eq. (9). The first one is the vaporization due to heat transfer from mixed gas (vapor side heat transfer): the first term of the RHS in Eq. (9), defined as $A_s(h_{gi}(T_g - T_i))/h_{fg}$. The second one is the vaporization due to heat transfer from liquid (liquid side heat transfer): the second term of the RHS in Eq. (9), defined as $A_s(-h_{li}(T_i - T_l))/h_{fg}$. The interface temperature T_i is calculated based on the instantaneous partial pressure of gaseous oxygen in the gas region ($T_i = T_{\text{sat}}(P_{\text{GOX}})$). The relative contribution differs with varying injection temperatures, flow rates and system pressures.

Fig. 10 shows the compared calculation results between 288 and 90 K helium injection. In the case of 90 K helium injection, the vapor side contribution is much lower than in the 288 K case, but due to the small partial pressure of the

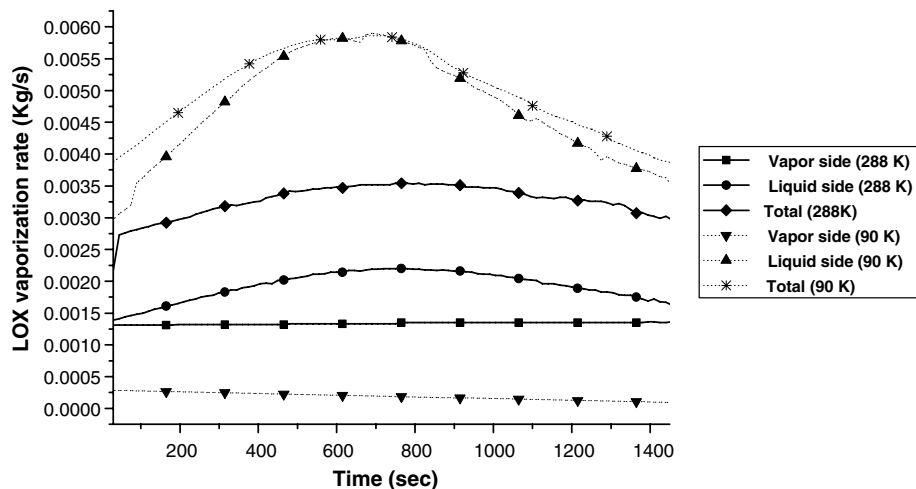


Fig. 10. Comparison between 288 K helium injection and 90 K helium injection.

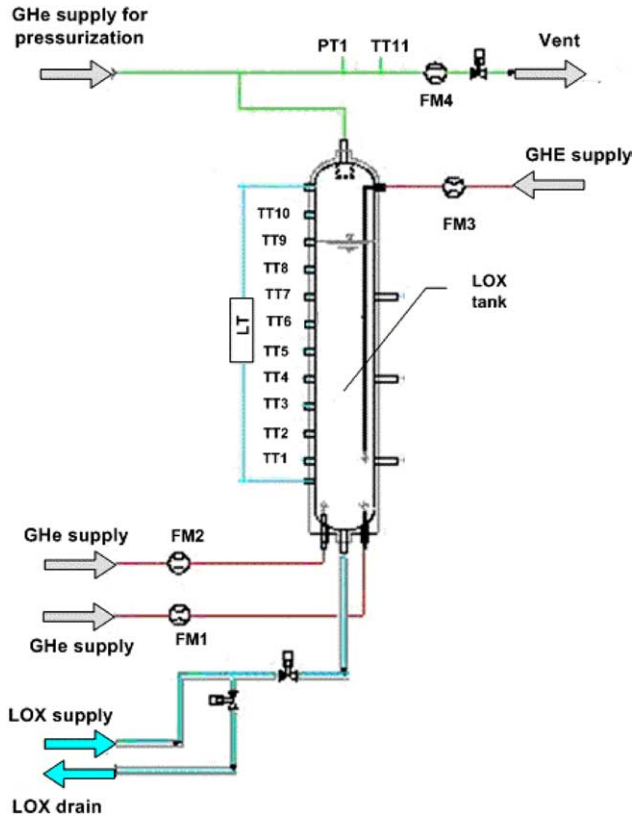


Fig. 11. Schematic diagram of test facility.

gaseous oxygen, the liquid side contribution is larger. In the case of 90 K injection, most of the heat transfer occurs by liquid side heat transfer. As shown in Fig. 10, the net vaporization rate for 90 K is larger than that for 288 K, which leads to more sub-cooling of the LOX. It is also noticeable that the vaporization due to the vapor side heat transfer can increase oxygen partial pressure in the gas region, which subsequently decreases the liquid side heat transfer driven by mass diffusion. Therefore, the vaporization rate ultimately decreases.

5. Experimental results and comparison with analysis

Fig. 11 shows the experimental setup for a 90 l, 2.5 m tall by 0.23 m diameter test tank. The tank is insulated with vacuum and multi-layer insulation. The axial temperature

distribution is measured with 10 RTD type temperature sensors. Helium can be injected at three ports as shown in the figure. The tank pressure can be controlled by atmospheric venting and a back pressure regulator installed at the vent port. Before the helium injection tests, ambient heating is calculated from the measured LOX temperature increase and the ullage pressure increments over time. Table 2 shows the tests and the analysis conditions with varying helium flow rates.

Fig. 12 shows the test and analysis results for varying injection flow rates under atmospheric condition (1 bar). As shown in the figure, a larger injection rate results in more cooling of the LOX. This is true because a larger helium injection increases the helium gas fraction in the bubbles and accordingly lowers the GOX fraction, which increases the liquid side heat transfer and net cooling. For 134 l/min ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 2.07 \text{ min}^{-1}$) injection rate, the measured LOX temperature reaches down to 85.3 K at 1500 s. The data obtained from two high injection rate cases ($\dot{V}_{\text{He}}/V_{\text{LOX}} = 1.39 \text{ min}^{-1}$, 2.07 min^{-1}) matches well.

Fig. 13 shows the sub-cooled temperature with two modified helium injection methods. To decrease initial bubble size, a porous nozzle (10 holes) is applied instead of a single nozzle, and to increase turbulent intensity, downward injection is applied instead of upward injection. When

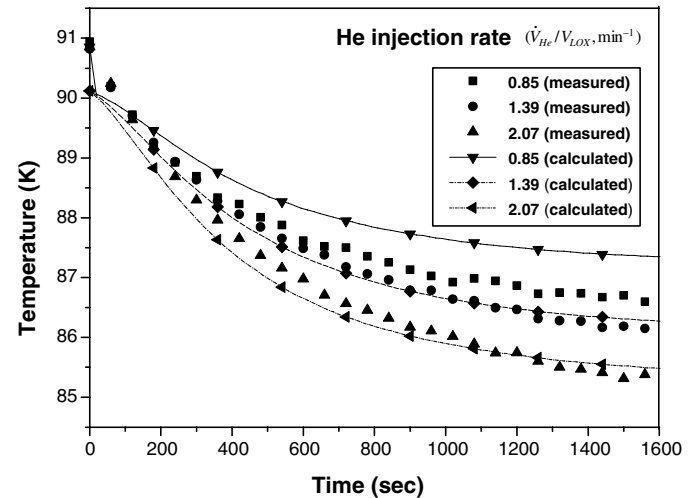


Fig. 12. Temperature sub-cooling with varying helium injection flow rates (atmospheric condition).

Table 2
Test and analysis conditions

Test and analysis conditions	Case 1	Case 2	Case 3
Helium injection rate (standard l/min) ($\dot{V}_{\text{He}}/V_{\text{LOX}}$ (l/min ⁻¹))	55.2 (0.85)	90.6 (1.39)	134.5 (2.07)
Filled LOX volume (l)	65.0	65.0	65.1
System pressure (atm)	1	1	1
Initial LOX temperature (K)	91.0	91.0	91.0
Ambient heating rate (W)	261	261	261
Average bubble diameter (mm)	4.7	4.8	5.3
Gas-to-interface heat transfer coefficient h_{gi} (W/m ² K)	41	40	41
Liquid-to-interface heat transfer coefficient h_{li} (W/m ² K)	315	330	338

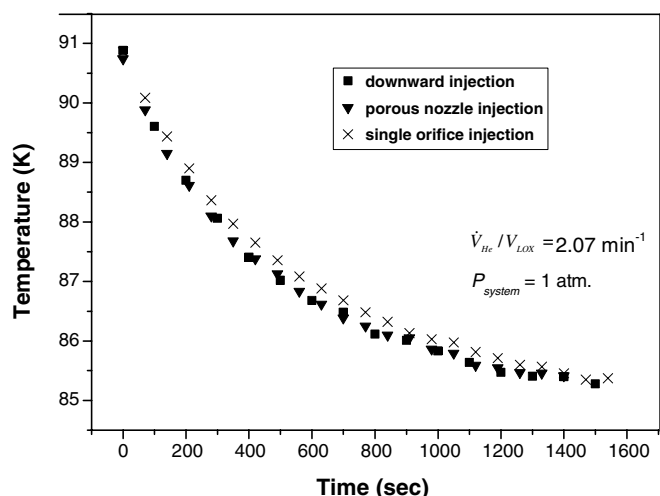


Fig. 13. Temperature sub-cooling with special helium injection methods.

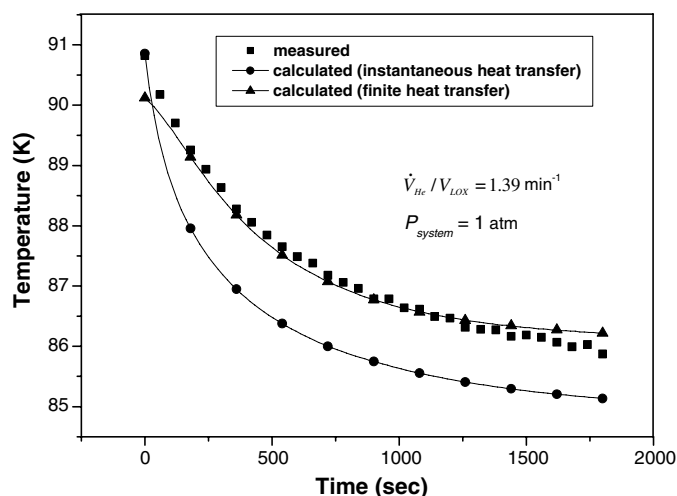


Fig. 14. Comparison of analytical results with test data.

the helium is injected in a downward direction, it may collide with the bottom wall, and the turbulence intensity may be increased. In the above two cases, 0.1–0.2 K improvement of cooling is obtained.

Fig. 14 shows the comparison of experimental and two analytical results. The curve with triangular marks is the case of finite heat transfer and instantaneous mass transfer, which is deeply discussed in this paper. The curve with circle marks is the case of instantaneous heat and mass transfer (equilibrium bubbler) for comparison. This curve actually indicates the ideal maximum sub-cooling of liquid oxygen. As shown in Fig. 14, the equilibrium bubbler model predicts the sub-cooled experimental temperature data with more than 1 K temperature difference. The case of finite-rate heat transfer and instantaneous mass transfer, however, shows much better agreement with the experimental data. The finite-rate heat transfer model looks very useful for helium injection application.

6. Conclusion

The characteristics of helium injection cooling into liquid oxygen chamber are thoroughly examined by theoretical, analytical and experimental methods. When the ratio of helium injection volume flow rate to the filled LOX volume is about 1 min^{-1} , the injection system can be treated as bubbly flow. Single bubble analysis can be extended to account for the whole bubbling system under these circumstances. The assumption of finite heat transfer and instantaneous mass transfer results in very satisfactory agreement with the experimental data. For the analysis of cooling rate, the heat transfer of the oxygen-liquid bath and the helium-gas bubble to the interface should be simultaneously considered. To maximize the cooling performance, enhancement of the heat transfer from liquid to interface is recommended. Incorporation of a porous nozzle to reduce bubble size, or changing bubble injection direction in the helium injection cooling system can only increase the sub-cooling temperature slightly.

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